

Alexandre Vérine

I am currently a Research Fellow at CSD at ENS PSL, specializing in Generative Models.

I defended my PhD in July 2024 at Université Paris-Dauphine under the supervision of Prof. Yann Chevalerey for which I received the Doctoral Grand Prize of the Chancellery of the Universities of Paris.

My research is centered around the expressivity of Generative Models and the trade-off between the quality and the diversity of the generated data.

alexverine.com

alexandre.verine@ens.fr

+33621340114

75 Boulevard Exelmans, 75016 Paris

EXPERIENCE	PR[AI]RIE - Junior AI Research Fellow <i>École Normale Supérieure PSL</i>	September 2024 - Today Paris, France
	PhD Student <i>Université Paris-Dauphine-PSL</i>	September 2020 - June 2024 Paris, France
AWARDS	Doctoral Grand Prize of the Chancellery of the Universities of Paris 2025 Prestigious award granted annually to recognize the best doctoral dissertations in the Paris region	Chancellery of the Universities of Paris
	Best Student Paper Quality/Diversity Trade-off in Generative Models	November 2023 Dauphine Digital Days 2023
	Best Student Paper Expressivity of Normalizing Flows	April 2022 LAMSADE Day 2022
PUBLICATIONS	When the temperature fails : Change the loss <i>Alexandre Vérine, Florian Le Bronnec, Kunhao Zheng, Benjamin Negrevergne, Yann Chevalerey Alexandre Allauzen</i> Forty-Second International Conference on Machine Learning, (ICML 2025)	
	Improving Discriminator Guidance in Diffusion Models <i>Alexandre Vérine, Ahmed Mehdi Inane, Florian Le Bronnec, Yann Chevalerey, Benjamin Negrevergne</i> European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML PKDD 2025)	
	Exploring Precision and Recall to assess the quality and diversity of LLMs <i>Alexandre Vérine, Florian Le Bronnec, Benjamin Negrevergne, Yann Chevalerey, Alexandre Allauzen</i> Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL 2024 Main)	
	Optimal Budgeted Rejection Sampling for Generative Models <i>Alexandre Vérine, Benjamin Negrevergne, Muni Sreenivas Pydi, Yann Chevalerey</i> The 27th International Conference on Artificial Intelligence and Statistics (AISTATS 2024)	

Precision-Recall Divergence Optimization for Generative Modeling with GANs and Normalizing Flows

Alexandre Vérine, Benjamin Negrevergne, Muni Sreenivas Pydi, Yann Chevalere
Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS 2023)

On the expressivity of bi-Lipschitz normalizing flows

Alexandre Vérine, Benjamin Negrevergne, Fabrice Rossi, Yann Chevalere
The 14th Asian Conference on Machine Learning (ACML 2022)

On the expressivity of bi-Lipschitz normalizing flows

Alexandre Vérine, Benjamin Negrevergne, Fabrice Rossi, Yann Chevalere
ICML Workshop on Invertible Neural Networks, Normalizing Flows, and Explicit Likelihood Models (INNF+2021)

PREPRINT

Spectral Collapse in Diffusion Inversion

Nicolas Bourrie, Alexandre Vérine, Auguste Genevesio

Learning To Sample From Diffusion Models Via Inverse Reinforcement Learning

Constant Bourdrez, Alexandre Vérine, Olivier Cappé

Equalized Generative Treatment: Matching f -divergences for Fairness in Generative Models

Alexandre Vérine, Rafael Pinot, Florian Le Bronnec

Conditional Distribution Quantization in Machine Learning

Blaise Delattre, Sylvain Delattre, Alexandre Vérine, Alexandre Allauzen

TALKS & More

Quality/Diversity Trade-off in Generative Models

Doctoral Conference AI for the Sciences

May 2024
Université PSL

Entre réalisme et inventivité

Popular Science Artical

March 2024
Dauphine Eclairages

Suming Up the Expressivity of Generative Models

TEDxDAuphine

February 2022
Université Paris-Dauphine-PSL

TEACHING

Generative AI

Université Paris-Dauphine-PSL - IASD Master's Degree

2025
Lectures

Introduction to Deep Learning

Université Paris-Dauphine-PSL - AI Bachelor

2024-2025
Lectures

Introduction to Deep Learning

Université Paris-Dauphine-PSL - Executive Master

2023-2025
Lectures

Trustworthy AI via Data Science Projects

Université Paris-Dauphine-PSL - Executive Master

2022-2024
Lectures

Machine Learning Projects

Université Paris-Dauphine-PSL - IASD Master's Degree

2022-2025
Lectures

Mathematics for Data Science

2020-2022

Université Paris-Dauphine-PSL - Master's Degree

Lectures/Seminars

Advanced Machine Learning - Normalizing Flow

2021

Université Paris-Dauphine-PSL - IASD Master's Degree

Lecture

**RESEARCH
INTERNSHIPS**

LAMSADE

September 2019 - June 2020

Université Paris-Dauphine-PSL

Paris, France

- Part-Time Research internship on generation of Adversarial Attacks with Invertible Neural Networks.

Machine Learning & Data Lab

April 2019 - August 2019

Wavestone

Paris, France

- Master's degree research internship on Invertible Neural Networks as a defense against Adversarial Attacks.

Advanced Structures & Composites Center

May 2018 - August 2018

University Of Maine

Orono, Maine, USA

- Research internship on organic photovoltaics materials. Developed a portable characterizing device for photovoltaic wire. Designed military application for the photovoltaic wire woven fabric.

EDUCATION

PhD in Artificial Intelligence

September 2020 - July 2024

Université Paris-Dauphine-PSL

Paris, France

- 3 years contract with LAMSADE Laboratory.
- Subject: Precision and Recall for Generative Models.
- Advisors: Yann Chevalere, Fabrice Rossi, Benjamin Negrevergne.

M.S Quantitative Economics

September 2019 - June 2020

Université Paris-Dauphine-PSL

Paris, France

- Last year of ENS Paris-Saclay as a multi-disciplinary one year program.
- Related Courses: Microeconomics, Macroeconomics, Econometrics, Game theory, Industrial Organization.

M.S. MVA - Mathematics, Vision & Learning

September 2018 – April 2019

École Normale Supérieure Paris-Saclay

Cachan, France

- Related courses: Reinforcement Learning, Deep Learning, Statistical Learning, Kernel Methods, Natural Language Processing, Astrophysics data processing, Probabilistic Graphical Models.
- Awarded with very high honors.

M.S. Electrical Engineering

September 2017 – May 2018

École Normale Supérieure Paris-Saclay

Cachan, France

- Related courses: Probabilities, Computing, Energy Processing, Signal Processing, Telecommunication, Automation.
- Research project: Thermic modelisation of a solar powered, self commuted, variable reluctance motor and life expectancy estimation for the french company SAUREA SAS.
- Awarded with high honors. Rank: 3/24.

M.S. Fundamental physics

September 2017 – September 2018

Université Paris-Sud

Orsay, France

- One year programm as evening lectures.
- Related courses: Plasma physics, atoms and molecule structure, atomic nucleus and particles, optical physics, laser physics.
- Awarded with high honors.

B.S. General Engineering September 2016 – September 2017
École Normale Supérieure Paris-Saclay Cachan, France

- Related courses: Mathematics, Computing, Mechanics, Energies, Numerical Electronics, Biologic electricity.
- Team Project: Building and designing an electronic spinet able to play any recorded song.
- Awarded with high honors. Rank: 16/60.

B.S General Engineering September 2014 - July 2016
Lycée Chaptal Paris, France

- Intensive 2-years course in preparation to sit the national competitive examinations for admission to the French Grandes Ecoles of physics and engineering.
- Related courses: Mathematics, Physics, Chemistry, Engineering, Computing.
- Individual Project: Building and designing the software, the hardware and the mechanical structure of a reduced SegWay System.